

Detection of genuine lossless audio files. Application to the MPEG-AAC codec. Detailed results.

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A. INTRODUCTION

This report is a complementary document for the paper submitted to the Journal of Audio Engineering Society: "Detection of genuine lossless audio files. Application to the MPEG-AAC codec." In this document, we give all detailed results, both on the small and large audio databases. Please refer to the paper for further details.

B. DETAILED RESULTS ON THE SMALL AUDIO DATABASE

A. Description of the database

The database is composed of 10 individual audio files, ripped from commercial CDs. The files are described in table I.

Id	Artist	Title	Album	Genre
Cocaine	J.J. Cale	Cocaine	Troubadour	Blues/Rock
Goldberg Variation n.10	J.S. Bach	Variation n.10	Goldberg Variations (G. Gould 1981)	Classical/piano
Evening	Paul Simon	Late in the Evening	One-Trick Pony	Pop/worldbeat
Fandango	Antonio Soler	Fandango	Variaciones del Fandango Español (A. Steier)	Classical/harpischord
Hold on	Johnny Copeland	Hold on to What You've Got	Jungle Swing	Blues
Messiah	G.F. Handel	And the Glory of the Lord	Messiah (W. Christie 1993)	Classical/vocal
Revolution	Tracy Chapman	Talkin' About Revolution	Tracy Chapman	Folk
Rose Rouge	Saint Germain	Rose Rouge	Tourist	Electro
Sentimental	Sonny Rollins	In a Sentimental Mood	Sonny Rollins with the Modern Jazz Quartet	Jazz
Symphony	L.v. Beethoven	1st movement - Allegro	6th Symphony (H.v. Karajan 1976)	Classical/symphony

TABLE I
DESCRIPTION OF THE AUDIO FILES IN THE SHORT DATABASE.

B. Distribution of relative scalefactors

In order to tune the detector, we analyze the distribution of relative scalefactors (i.e. scalefactors normalized by the dead-zone upper limit in each subband) in AAC encoded bitstreams, for each file in the database and for each bitrate: 128, 256 and 320 kbit/s. The distributions are normalized (i.e. sum of distribution values equals 1), and plotted as a function of the relative scalefactor defined in $[0, 1]$. We also plot (in red) the mean value.

On figure 1, we plot the overall distribution, averaged over all files.

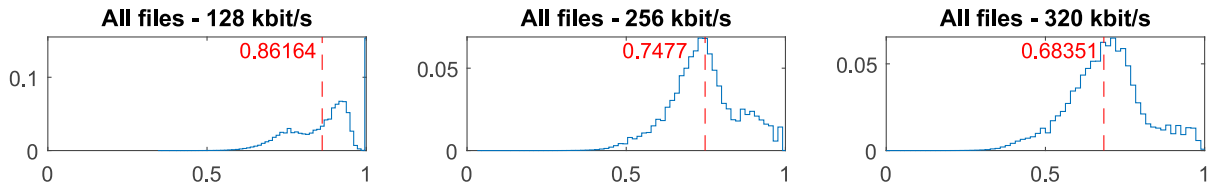


Fig. 1. Distribution of relative scalefactor in AAC encoded bistreams, all files.

On figure 2 and 3, we plot the distribution for each file.

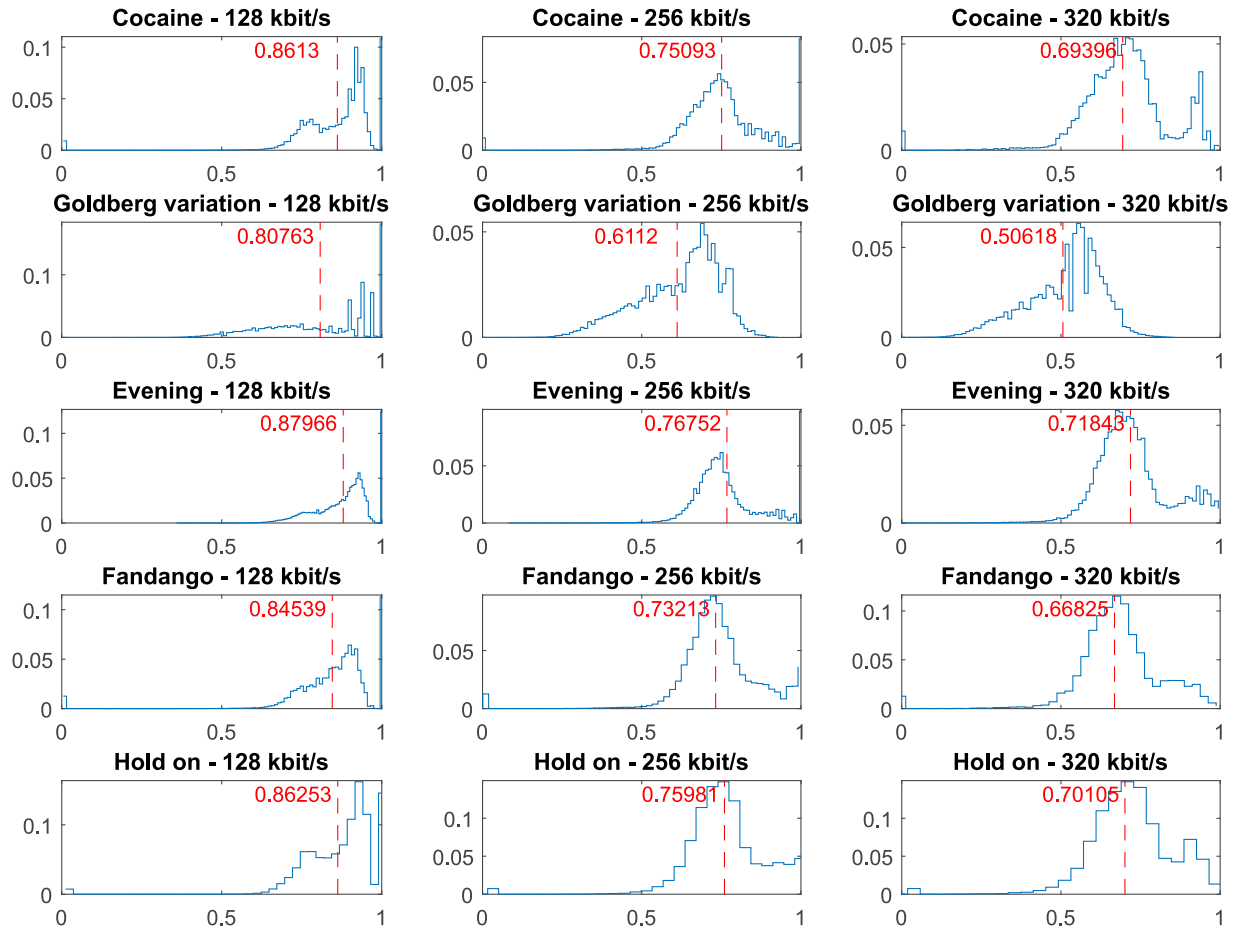


Fig. 2. Distribution of relative scalefactor in AAC encoded bistreams, part I.

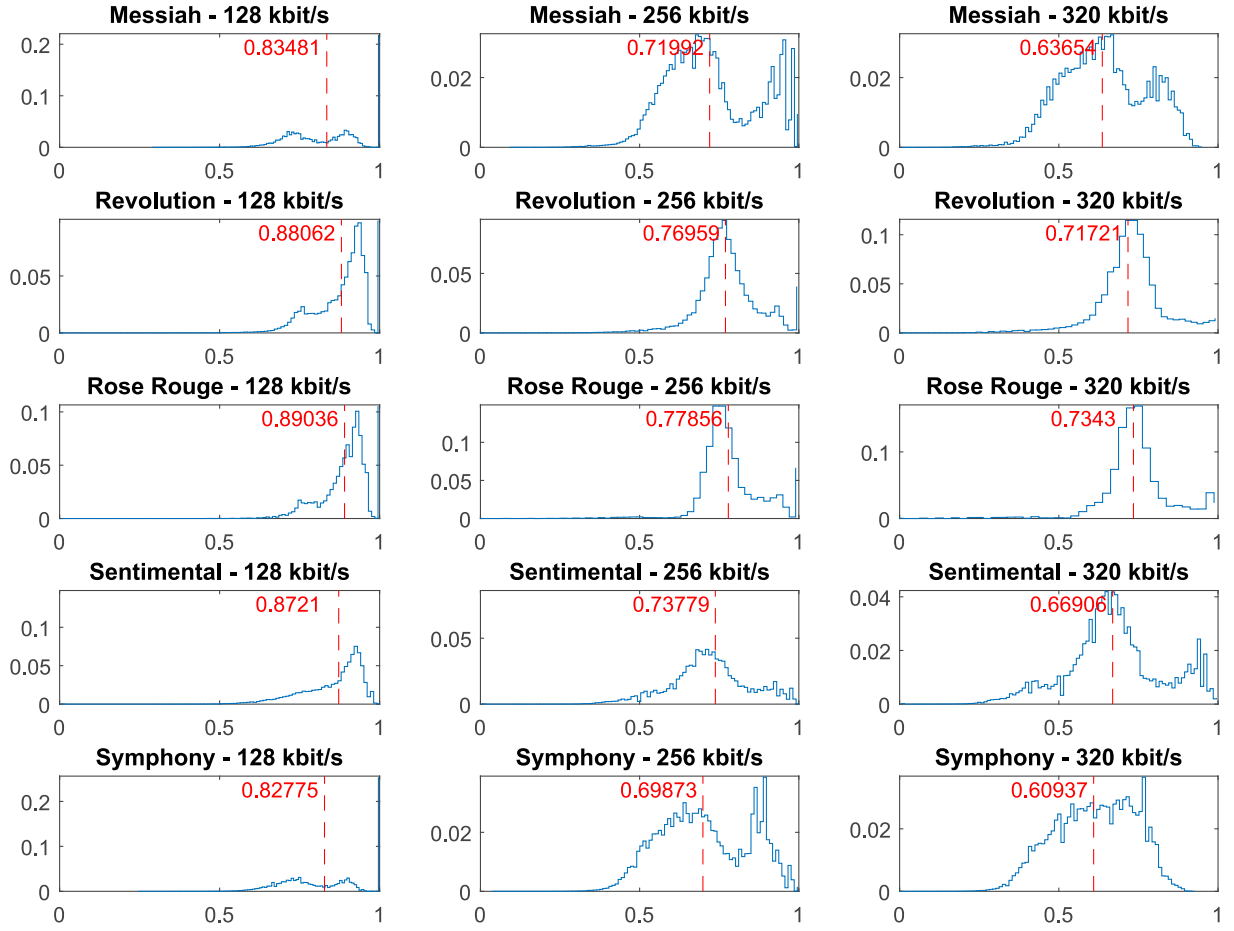
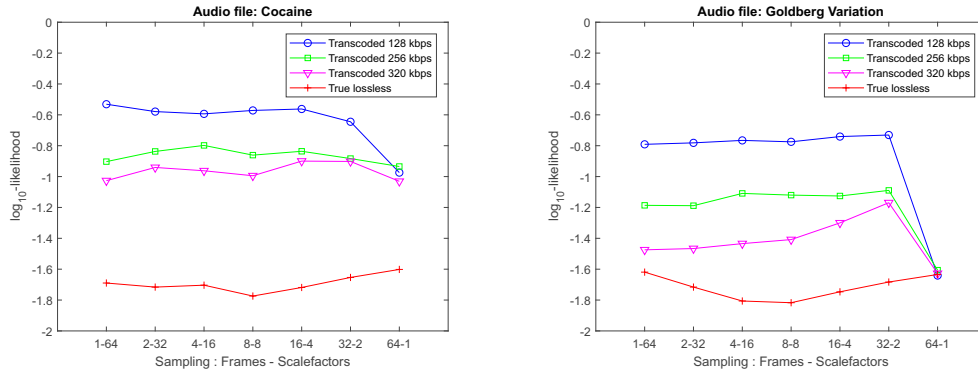


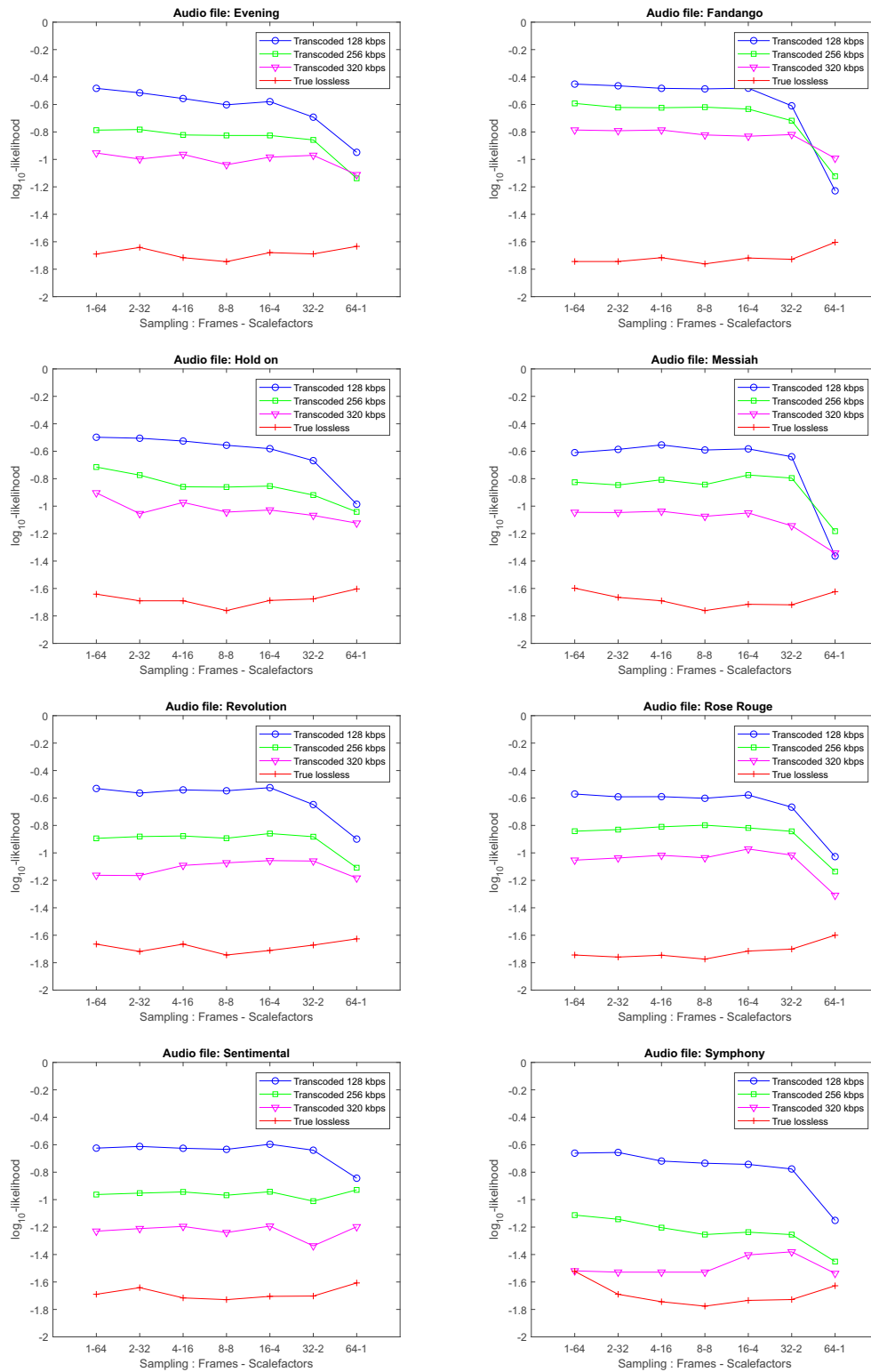
Fig. 3. Distribution of relative scalefactor in AAC encoded bistreams, part II.

C. Detection results

In order to set the optimal values for the frame and scalefactor samplings, we apply the detection algorithm to unprocessed files (corresponding to the *true lossless* case) and to coded-decoded files using MPEG AAC-LC (from iTunes software) operating at 128, 256 and 320 kbit/s (corresponding to the *transcoded* case).

On figure 4 and 5, we plot the log-likelihood values for 64-point sampling (frames $\times$ scalefactors).

Fig. 4. Log-likelihood values, 64-point sampling (frames $\times$ scalefactors), part I.

Fig. 5. Log-likelihood values, 64-point sampling (frames $\times$ scalefactors), part II.

C. DETAILED RESULTS ON THE LARGE AUDIO DATABASE

A. Description of the database

The database is composed of 100 albums (double or triple CDs are counted as one), from various genres, ripped from commercial CDs. The database is described in tables II and III.

Id	Artist	Title	Genre	Tracks	Company
1	B.B. King	Let the Good Times Roll	Blues	18	MCA
2	Big Bill Broonzy	Black, Brown & White (best of)	Blues	13	Black and Blue
3	Eric Clapton	Unplugged	Blues	14	Reprise
4	Johnny Copeland	Jungle Swing	Blues	11	Verve
5	J.J. Cale	Troubadour	Blues/rock	12	Mercury
6	Stevie Ray Vaughan	Couldn't Stand the Weather	Blues/rock	13	Epic / Legacy
7	Tony Joe White	Closer to the Truth	Blues/rock	12	Mercury
8	ZZ Top	Eliminator	Blues/rock	11	Warner Bros
9	Air	Pocket Symphony	Electro	12	Parlophone / Warner
10	Jean-Michel Jarre	Oxygène	Electro	6	Dreyfus
11	St Germain	Tourist	Electro	9	Blue Note
12	Thievery Corporation	The Cosmic Game	Electro	16	Ada Global
13	Niel Young	Harvest	Folk	10	Reprise
14	Simon and Garfunkel	The Concert in Central Park	Folk	19	Geffen
15	Suzanne Vega	Solitude Standing	Folk	11	A&M
16	Tracy Chapman	Tracy Chapman	Folk	11	Elektra / Asylum
17	Buena Vista Social Club	Buena Vista Social Club	Cuban folk	11	World Circuit
18	Lhasa	La Llorona	Pop/worldbeat	11	WEA
19	Idir	A Vava Inouva	Algerian folk	17	Globe Music
20	Värtinä	Ilmatar	Finnish folk	11	BMG
21	The Blues Brothers	Complete Recordings	Rhythm'n blues	33	Atlantic
22	Wilson Pickett	The Wicked Pickett	Rhythm'n blues	12	Atlantic
23	Otis Redding	The Dock of the Bay (best of)	Soul	20	Atlantic
24	Ray Charles	The Very Best of	Soul	15	Rhino
25	James Brown	Hot on the One	Funk	13	Polydor
26	Aretha Franklin	The Best of	Soul	12	Atlantic
27	Marvin Gaye	The Very Best of	Soul	22	Motown
28	Stevie Wonder	Original Musiquarium I (best of)	Funk	16	Motown
29	Julian "Cannonball" Adderley	Somethin' Else	Jazz	6	Blue Note
30	Chet Baker	Chet	Jazz	10	Fantasy Concord
31	John Coltrane	Stellar Regions	Jazz	11	Impulse
32	Miles Davis	Kind of Blue	Jazz	5	Columbia / Legacy
33	Duke Ellington	Ellington at Newport	Jazz	20	Sony Music
34	Bill Evans	The Paris Concert Edition One	Jazz	9	Universal
35	Richard Galliano	New York Tango	Jazz	11	BMG
36	Glenn Miller	Original Recordings (best of)	Jazz	15	RCA
37	Wes Montgomery	Full House	Jazz	9	Hallmark
38	Django Reinhardt	The Quintessence 1934-1943	Jazz	36	Fremaux
39	Nina Simone	Me me quite pas (best of)	Jazz	22	Mercury
40	Art Tatum	Over the Rainbow	Jazz	21	Sony Music
41	The Commitments	The Commitments vol.1	Movie soundtrack	14	Geffen
42	Various	Pulp Fiction	Movie soundtrack	13	Universal
43	Various	Sweet and Lowdown	Movie soundtrack	15	SBA
44	Ennio Morricone	Once Upon a Time in America	Movie soundtrack	15	SBA
45	Chuck Berry	Greatest hits (best of)	Rock'n Roll	18	Universal
46	Fats Domino	Original Recordings (best of)	Rock'n roll	26	Capitol
47	Buddy Holly	My Greatest Songs (best of)	Rock'n roll	14	Geffen
48	Jerry Lee Lewis	Original recordings (best of)	Rock'n roll	22	Mis
49	Eric Clapton	461 Ocean Boulevard	Pop/rock	10	Polydor

TABLE II
DESCRIPTION OF THE CDs IN THE LONG DATABASE - PART I.

Id	Artist	Title	Genre	Tracks	Company
50	Creedence Clearwater Revival	Willy and the Poor Boys	Pop/rock	10	Fantasy Concord
51	Dire Straits	Alchemy	Pop/rock	11	Mercury
52	The Eagles	Hotel California	Pop/rock	9	Elektra / Asylum
53	Midnight Oil	Diesel and Dust	Pop/rock	11	Columbia
54	Pink Floyd	The Wall	Pop/rock	26	EMI
55	The Police	Every Breath You Take (best of)	Pop/rock	13	A&M
56	R.E.M.	Out of Time	Pop/rock	11	Universal
57	Radiohead	OK Computer	Pop/rock	12	XI Recordings
58	Rammstein	Rosenrot	Hard rock	11	Universal
59	Bruce Springsteen	Greatest Hits vol.1 (best of)	Pop/rock	18	Columbia
60	Chris Rea	The Road to Hell	Pop/rock	10	East West
61	U2	The Joshua Tree	Pop/rock	11	Mercury
62	The Who	Who's Next	Pop/rock	16	Polydor
63	The Beatles	1962-1966 (Remaster)	Pop/rock	26	Calderstone
64	Doves	The Last Broadcast	Pop/rock	12	EMI
65	Peter Gabriel	So	Pop/rock	9	Real World
66	Genesis	Invisible Touch	Pop/rock	8	Parlophone
67	Roxy Music	Avalon	Pop/rock	10	Virgin
68	Steely Dan	Then and Now (best of)	Pop/rock	16	Geffen
69	Sting	The Soul Cages	Pop/Rock	9	A&M
70	Supertramp	The Very Best vol.1 and 2	Pop/rock	29	A&M
71	J.S. Bach	Goldberg Variations (G. Gould 1981)	Classical/piano	32	Sony Classical
72	L.v. Beethoven	Fazil Say plays Beethoven (F. Say)	Classical/piano	9	Naive
73	Frédéric Chopin	Nocturnes (D. Barenboim)	Classical/piano	21	Deutsche Grammophon
74	Edvard Grieg	Lyric Pieces (E. Guilels)	Classical/piano	20	Deutsche Grammophon
75	W.A. Mozart	Piano Sonatas 3, 7 & 11 (C. Zacharias)	Classical/piano	9	EMI
76	Frantz Schubert	Quintet "Die Forelle" (C. Zacharias)	Classical/piano	10	MDG
77	Antonio Vivaldi	Chamber Concertos (Il Giardino Armonico)	Classical/orchestra	75	Teldec
78	G.F. Handel	Water Music, Royal Fireworks (J.C. Malgoire)	Classical/orchestra	22	Sony Classical
79	G.P. Telemann	Orchestral Suites (Akademie für Alte Music)	Classical/orchestra	32	Harmonia Mundi
80	J.P. Rameau	Hyppolite et Aricie (S. Kuijken)	Classical/orchestra	27	Deutsche Harmonia Mundi
81	J.S. Bach	Brandenburg Concertos (N. Harnoncourt)	Classical/orchestra	19	Teldec
82	J.C.C. de Mondonville	6 Sonatas (M. Minkowski)	Classical/orchestra	19	Brilliant
83	Felix Mendelsson	Trios for Piano, Violin and Cello (I. Stern)	Classical/orchestra	8	Sony Classics
84	W.A. Mozart	Eine Kleine Nachtmusic (N. Harnoncourt)	Classical/orchestra	10	Teldec
85	P. Tschaikowsky / F. Mendelsson	Violin concertos (C. Abbado)	Classical/orchestra	6	Deutsche Grammophon
86	Bedrich Smetana	My Fatherland (R. Kubelik)	Classical/orchestra	6	Deutsche Grammophon
87	Marin Marais	Suites for Viola da Gamba (J.M. Quintana)	Classical/viola	27	Harmonia Mundi
88	J.S. Bach	Suites for Cello (P. Tortelier)	Classical/cello	36	EMI
89	L.v. Beethoven	Symphonies 2 & 7 (H.v. Karajan 1976)	Classical/symphony	8	Deutsche Grammophon
90	W.A. Mozart	Symphonies 25 & 29 (L. Bernstein)	Classical/symphony	11	Deutsche Grammophon
91	Felix Mendelsson	Symphonies 4 & 5 (N. Marriner)	Classical/symphony	11	Deutsche Grammophon
92	Gustav Holst	The Planets (H.v. Karajan)	Classical/symphony	7	Deutsche Grammophon
93	Frantz Schubert	Symphonies 8 & 9 (G. Szell)	Classical/symphony	6	Sony Classics
94	N. Rimsky-Korsakov	Sheherazade (E. Ormandy)	Classical/symphony	10	Sony Classics
95	Edvard Gieg	Peer Gynt (H. Bloomstedt)	Classical/vocal	19	Decca
96	W.A. Mozart	Mass in C minor (P. Herreweghe)	Classical/vocal	13	Harmonia Mundi
97	Carl Orff	Carmina Burana (E. Ormandy)	Classical/vocal	26	Sony Classics
98	G.F. Handel	Messiah (W. Christie)	Classical/vocal	38	Harmonia Mundi
99	W.A. Mozart	Die Zauberflöte, highlights (K. Böhm)	Classical/vocal	16	Deutsche Grammophon
100	Various	Arie Antiche (C. Bartoli)	Classical/vocal	21	Decca

TABLE III
DESCRIPTION OF THE CDS IN THE LONG DATABASE - PART II.

B. Detection results

In this section, we give the final classifications results for stereo AAC-*transcoded* files at 256 kbit/s and at 320 kbit/s. Classification results for *true lossless* and *transcoded* at 128 kbit/s are not plotted, since the classification is correct for all files and all settings. The detection threshold is always chosen such that the false positive ratio is 0.

On figures 6 and 7, we plot the results for 8x8 setting, and detection threshold $\lambda = 0.031$. Total false negative ratio is 1.9% at 256 kbit/s and 3.9% at 320 kbit/s.

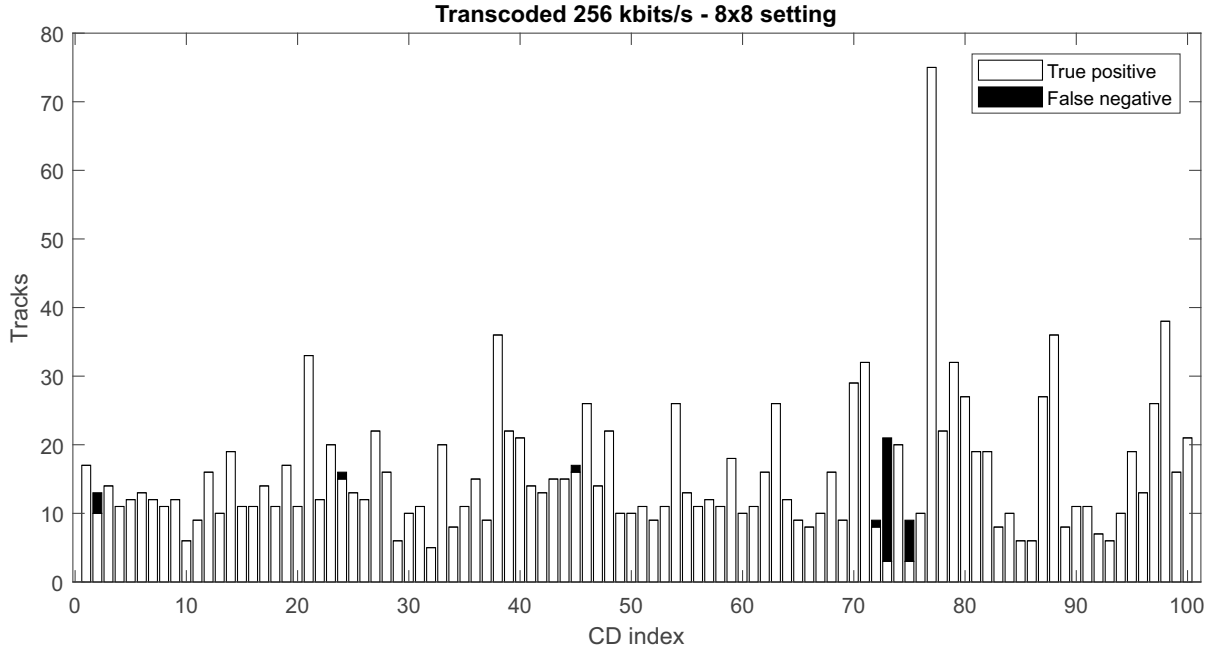


Fig. 6. Number of true positives and false negatives on *transcoded* audio files, for each CD in the database, $\lambda = 0.031$.

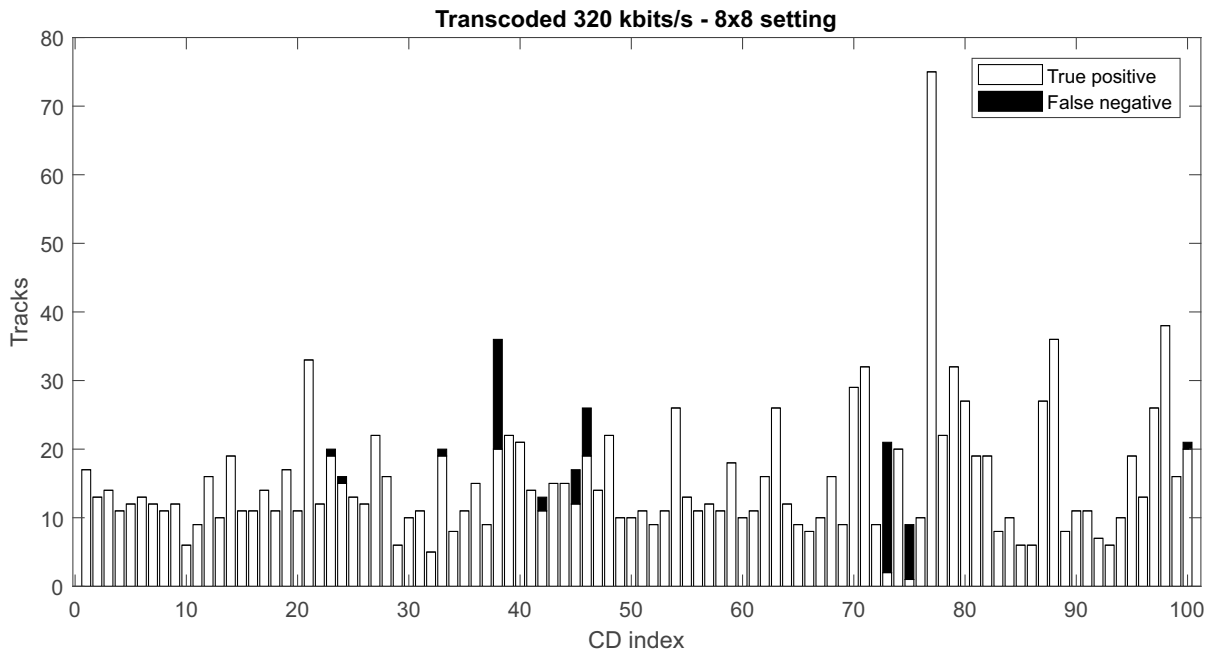


Fig. 7. Number of true positives and false negatives on *transcoded* audio files, for each CD in the database, $\lambda = 0.031$.

On figures 8 and 9, we plot the results for 16x16 setting, and detection threshold $\lambda = 0.019$. Total false negative ratio is 0.4% at 256 kbits/s and 1.1% at 320 kbit/s.

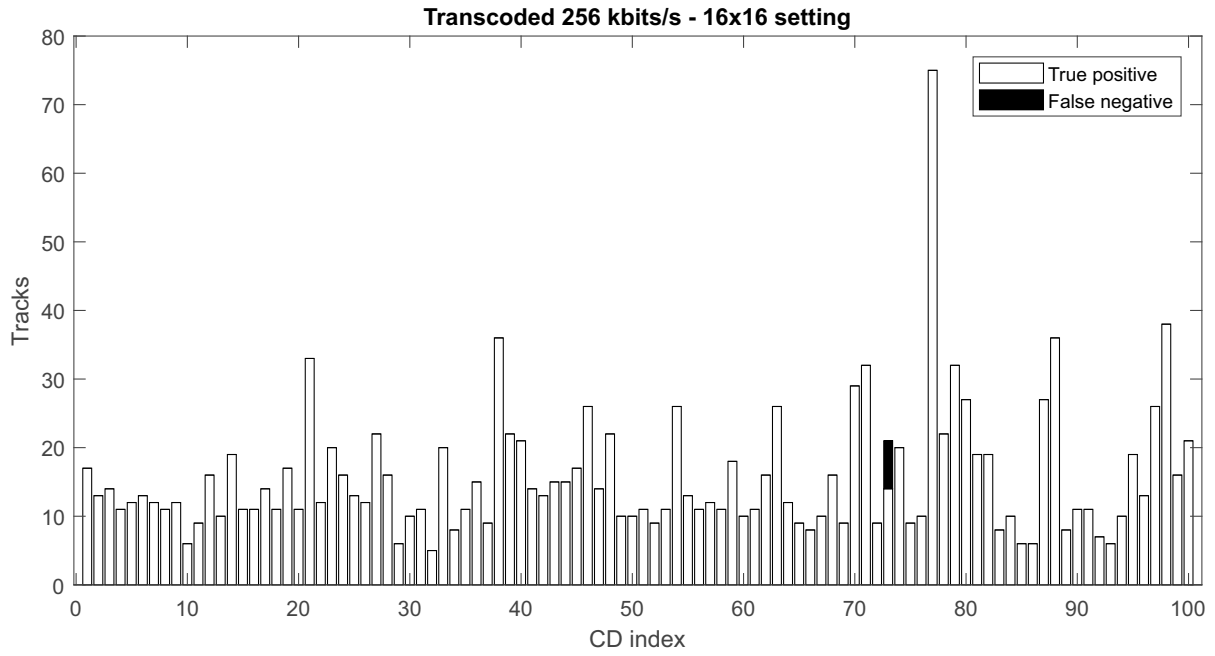


Fig. 8. Number of true positives and false negatives tracks for each CD in the database, $\lambda = 0.019$.

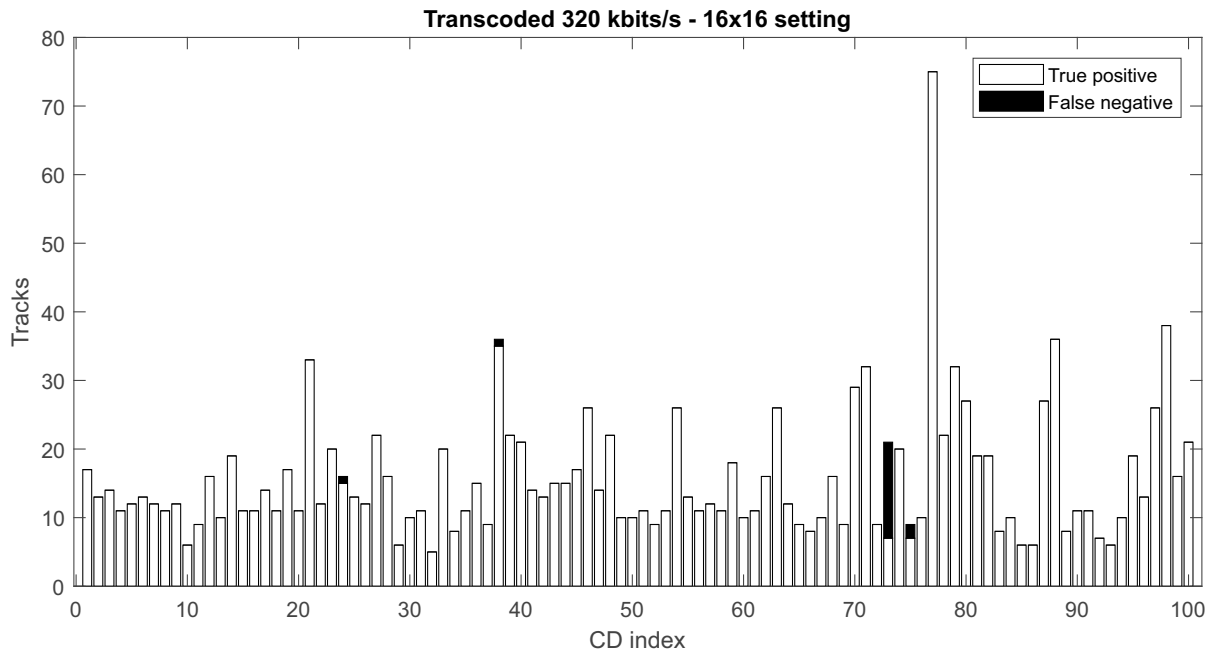


Fig. 9. Number of true positives and false negatives tracks for each CD in the database, $\lambda = 0.019$.

On figures 10 and 11, we plot the results for 32x32 setting, and detection threshold $\lambda = 0.0145$. Total false negative ratio is 0.06% at 256 kbits/s and 0.3% at 320 kbit/s.

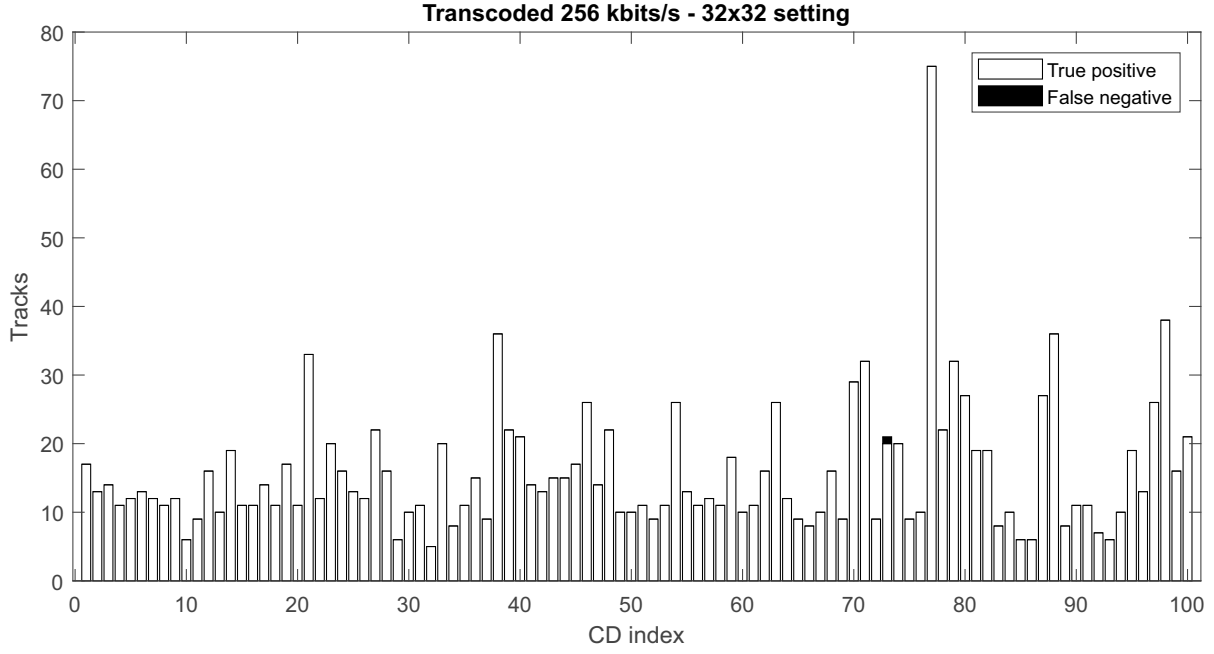


Fig. 10. Number of true positives and false negatives tracks for each CD in the database, $\lambda = 0.0145$.

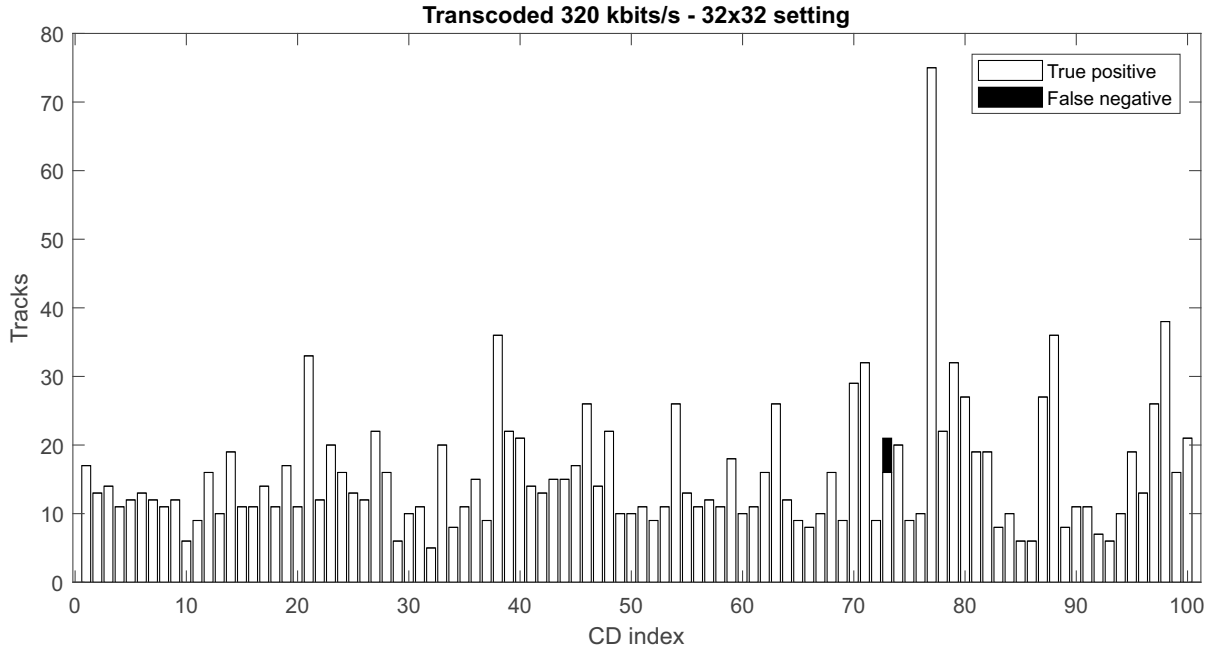


Fig. 11. Number of true positives and false negatives tracks for each CD in the database, $\lambda = 0.0145$.

For the 64x64 setting and $\lambda = 0.0125$, false positive and false negative ratios are 0 both at 256 kbit/s and 320 kbit/s.